

Course Code	Name of the Course				
216U03C305	Electrical Networks				
Teaching Scheme	TH	P	TUT	Total	
	03	-	01	04	
Credits Assigned	03	-	01	04	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	25	20	30	50	125

Course pre-requisites:

- Elements of Electrical and Electronics Engineering
- Partial fraction expansion, matrices, Laplace transforms and differential equations

Course Objectives:

Electrical Networks find its applications in various allied disciplines of Electronics Engineering and Telecommunication Engineering. This course will help the students to learn the fundamental concepts of mesh and nodal analysis, network theorems and functions, transient analysis, two port analysis, synthesis of realizable circuits from transfer functions and passive filters.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

- CO1.** Analyse DC and AC circuits using mesh analysis, nodal analysis and network theorems.
- CO2.** Analyse transient and steady state response in time and frequency domain.
- CO3.** Determine two port network parameters.
- CO4.** Synthesize circuits from transfer functions.
- CO5.** Learn and design different types of filters and their parameters

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs.	CO
1	Network Analysis		14	CO1
	1.1	DC with Dependant Sources: Mesh and nodal analysis, Superposition theorem, Thevenin's, Norton's Theorem and Maximum Power Transfer Theorem		
	1.2	AC Circuits: Mesh and nodal analysis, Superposition theorem, Thevenin's theorem, Norton's Theorem and Maximum Power Transfer Theorem		
	1.3	Coupled Circuits: Self and mutual inductance, Coefficient of coupling, equivalent inductances and mesh analysis using dot convention.		
2	Transient response of R-L, R-C, R-L-C circuits (Series and Parallel combination)		10	CO2
	2.1	Test inputs for transient response: step, impulse and ramp input.		
	2.2	Initial conditions		
	2.3	Solution based on time domain analysis		
	2.4	Solution based on Laplace transform		
3	Two port networks (Z,Y, ABCD and Hybrid Parameters)		08	CO3
	3.1	Driving point and transfer function		
	3.2	Concept of reciprocity and symmetry		
	3.3	Two port network parameters and their interrelation		
	3.4	Interconnection of two port networks: Cascade connection in terms of ABCD parameters.		
4	Network functions and Synthesis		08	CO4
	4.1	Poles and Zeros of network functions and their significance		
	4.2	Time domain response using pole zero plot		
	4.3	Hurwitz polynomial		
	4.4	Positive real functions		
	4.5	Properties and Synthesis of LC circuit using Foster I and II, Cauer I and II		
5	Passive Filters		05	CO5
	5.1	Classification of Filters: Low pass, high pass, band pass and band stop filters, transfer functions, frequency response and filter parameters.		
	5.2	T and pi network		
	5.3	Design of constant k filters: Low pass, high pass, band pass and band stop filters		
	5.4	Design of constant m-derived filters: Low pass, high pass, band pass and band stop filters		
Total			45	

Reference Books

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Ravish Singh	<i>Network Analysis and Synthesis</i>	McGraw Hill Education (India) Private Limited.	Seventh Reprint 2018
2	D. Roy Choudhury	<i>Networks and Systems</i>	New Age International Publishers.	2 nd Edition 2010
3	A. Sudhakar and Shyammohan. S. Palli	<i>Circuits and Networks</i>	McGraw Hill Education (India) Private Limited.	4 th Edition 2010
4	C. K. Alexander and M. N. O. Sadiku	<i>Electric Circuits</i>	McGraw Hill Education (India) Private Limited.	5 th Edition 2013
5	Nilsson Riedel	<i>Electric Circuits</i>	Pearson Education India.	11 th Edition 2018

Instructions:

- TUT includes journal / written record of tutorial .Tutorials will be evaluated out of **25 marks each** A minimum of 8-10 tutorials will be performed in a semester. Average of all tutorials will be considered as final TUT CA marks.